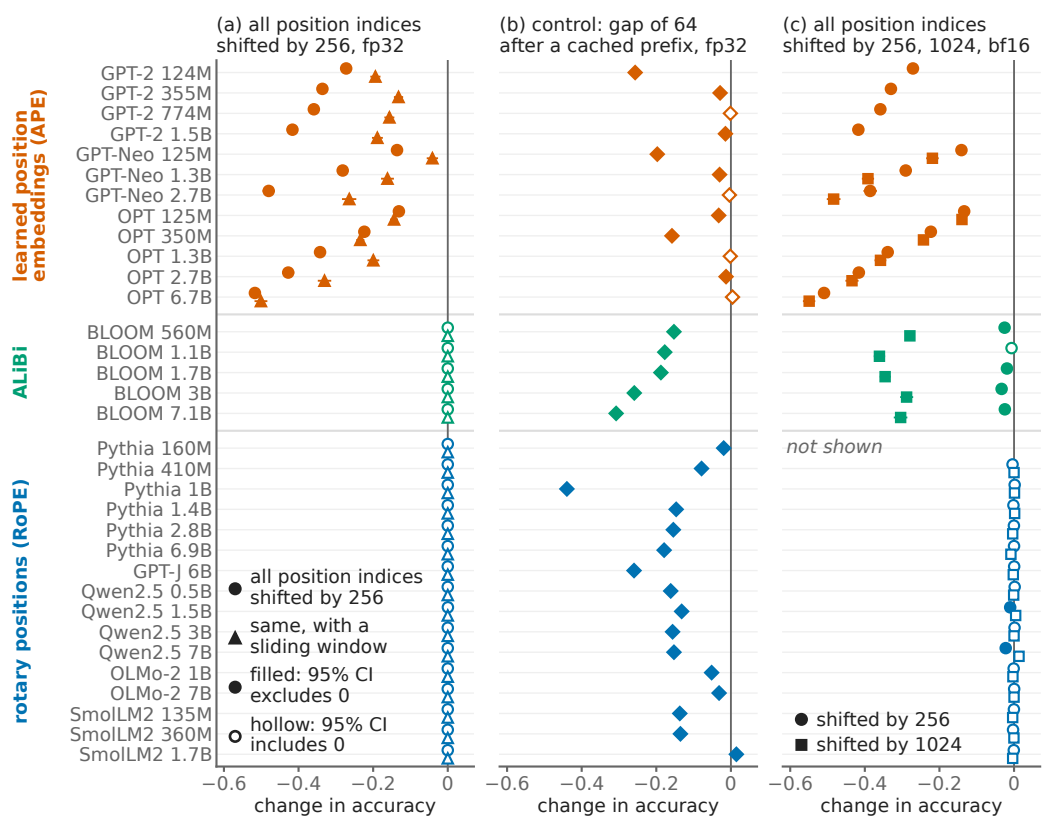
(a) all position indices shifted by 256, fp32
(b) control: gap of 64 after a cached prefix, fp32
(c) all position indices shifted by 256, 1024, bf16
learned position embeddings (APE)
ALiBi
rotary positions (RoPE)
GPT-2 124M
GPT-2 355M
GPT-2 774M
GPT-2 1.5B
GPT-Neo 125M
GPT-Neo 1.3B
GPT-Neo 2.7B
OPT 125M
OPT 350M
OPT 1.3B
OPT 2.7B
OPT 6.7B
BLOOM 560M
BLOOM 1.1B
BLOOM 1.7B
BLOOM 3B
BLOOM 7.1B
Pythia 160M
Pythia 410M
Pythia 1B
Pythia 1.4B
Pythia 2.8B
Pythia 6.9B
GPT-J 6B
Qwen2.5 0.5B
Qwen2.5 1.5B
Qwen2.5 3B
Qwen2.5 7B
OLMo-2 1B
OLMo-2 7B
SmolLM2 135M
SmolLM2 360M
SmolLM2 1.7B
all position indices shifted by 256
same, with a sliding window
filled: 95% CI excludes 0
hollow: 95% CI includes 0
not shown
shifted by 256
shifted by 1024
−0.6
−0.4
−0.2
0
change in accuracy